Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

For layered thin films, the central structural question is often not only which phase is present, but how the crystallites are oriented relative to the substrate. In layered van der Waals solids and related two-dimensional materials, this orientational state can directly control the properties that motivate the films in the first place.[1–3] In layered halides such as PbI_2, transport and device response can depend strongly on whether the layers lie parallel to the substrate.[4] In topological bismuth chalcogenides, strong c-axis alignment is needed to preserve the desired electronic response.[5] In transition-metal dichalcogenides, catalytic activity can increase when platelet edges are exposed by the film orientation.[6] Comparable orientational states also occur across a wider set of layered thin-film families, including telluride chalcogenides (e.g., Bi_2Te_3, Sb_2Te_3), dichalcogenides (e.g., MoS_2, NbS_2), layered halides and oxyhalides (e.g., BiI_3, BiOI), and laminated 2D-flake films such as $Ti_3C_2T_x$ MXenes and graphene-oxide papers.[7–14]

Wide-angle x-ray scattering (WAXS) with an area detector is attractive for this problem because a single exposure probes a large reciprocal-space volume rich with information of both structure and orientation.[15–19] The difficulty is that many layered films do not fall cleanly into either of the standard diffraction limits. They are neither single crystals, where reciprocal space consists of discrete points, nor fully isotropic three-dimensional (3D) powders, where orientational averaging produces spherical shells.[20] Instead, a common thin-film orientational state is the two-dimensional (2D) oriented powder, in which plate-like crystallites share an aligned layer normal while remaining azimuthally randomized in-plane.[3, 20, 21] This intermediate orientational state produces detector patterns that inherit features of both limiting cases while necessitating nontraditional analysis techniques.[20]

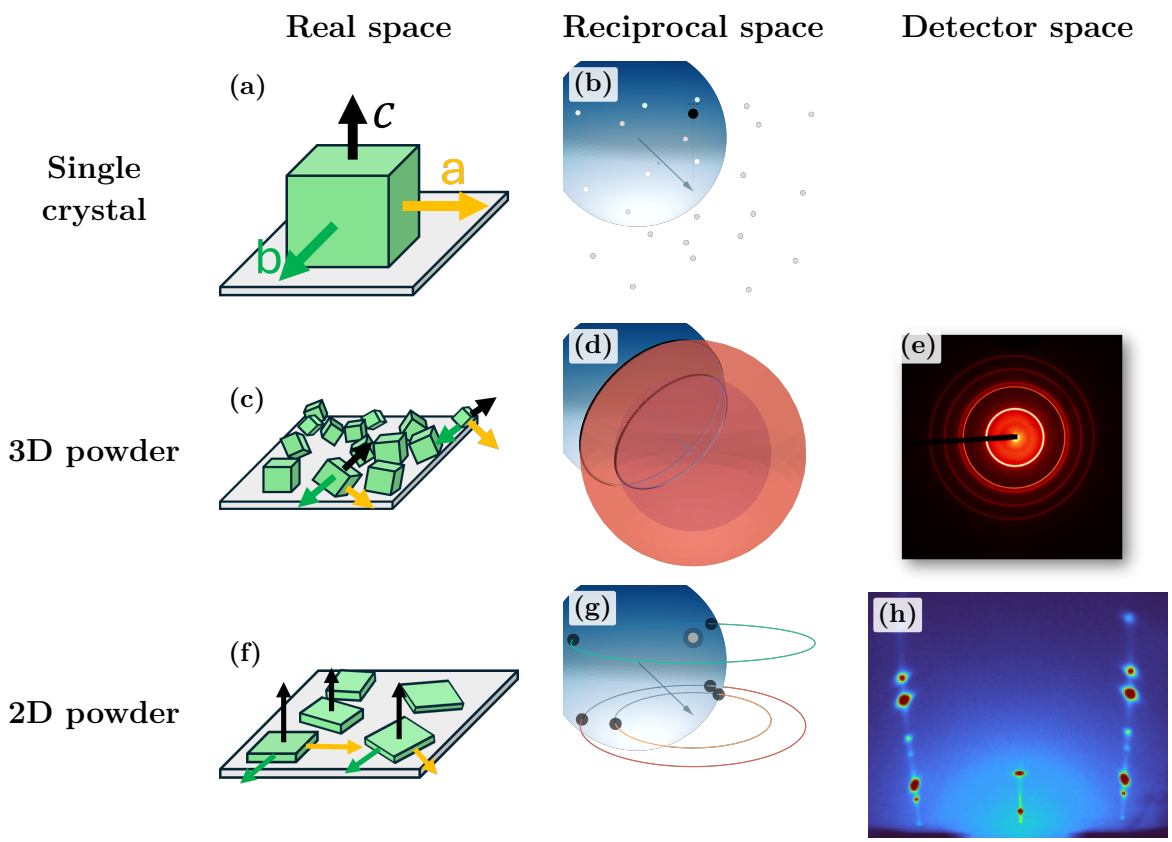

Figure 1: Diffraction across three orientational limits: single crystal, 3D powder, and 2D powder. Columns show the corresponding real-space, reciprocal-space, and detector-space views, while rows group the orientational limits. Panels (a,b) show a single crystal in real space and reciprocal space. Panels (c–e) show a 3D powder, with randomly oriented crystallites, spherical reciprocal-space shells, and Debye–Scherrer rings on the detector. Panels (f–h) show a 2D powder, with aligned surface normals and in-plane orientational disorder, reciprocal-space rings with specular Bragg features, and the corresponding detector pattern with symmetry about the central axis.

Figure 1 summarizes the real-space, reciprocal-space, and detector-space consequences of these three orientational limits. In the Ewald construction, elastic scattering requires that reciprocal-space intensity lie on the Ewald sphere of radius $|\mathbf{k}|$.[22] For a single crystal, reciprocal space consists of isolated Bragg points (Fig. 1b), so at fixed incidence angle the Ewald sphere typically intersects only a small subset of reflections and the detector records sparse spots.[22] For a 3D powder, random crystallite orientations smear the intensity of each Bragg point over a spherical shell, and the Ewald-sphere intersection produces Debye–Scherrer rings on the detector (Fig. 1d, Fig. 1e).[22] For a 2D powder, azimuthal disorder about the common c axis smears each off-specular Bragg point into an in-plane ring about Q_z, while specular $(00L)$ reflections remain localized near the Q_z axis (Fig. 1g).[15, 20] The corresponding illustration in Fig. 1f schematically shows 2D platelets with aligned growth, and a representative 2D-powder detector pattern is shown in Fig. 1h. When the Ewald sphere intersects this reciprocal-space distribution, the detector records hybrid signatures: pairs of symmetric spots from the off-specular ring intersections and single-crystal-like features near

the specular stripe from $(00L)$ intensity.[15, 17, 23] Mosaicity then redistributes intensity, broadening the ring intersections into arcs and spreading the $(00L)$ features into cap-like intensity near the specular region.[15, 20]

This hybrid detector-space information is what makes the 2D-powder problem both rich and difficult. Many standard WAXS workflows reduce area-detector images to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[17, 19, 24–26] Those steps suit nearly isotropic powders, but for 2D powders they deweight or remove off-specular arcs and specular caps that constrain out-of-plane correlations, stacking, and mosaic tails.[15, 17] Stacking disorder is also commonly inferred from 1D interference models and refinements,[27–29] which do not directly exploit the two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with orientation-distribution-function (ODF) refinements or preferred-orientation corrections,[30–35] but those approaches are more general than needed for the narrowly constrained 2D-powder case considered here.[34, 35] Recent radial-profile approaches have made thin-film GIXD substantially more quantitative, but peak overlap and background separation can remain limiting in strongly anisotropic patterns, especially at small incidence angles.[36] For the layered films considered here, the measured morphology already motivates axial symmetry about the film normal and a low-dimensional out-of-plane mosaicity kernel.[3, 20, 21] We therefore assume this physically motivated 2D-powder orientation distribution a priori and refine it with a detector-space forward model, using off-specular arcs and specular-family line shapes as direct constraints.[15, 17]

Here we present a detector-space forward model for 2D-powder WAXS. Starting from an indexed crystallographic model, it predicts area-detector images and applies the same projection and integration procedures to measured and calculated intensity. The comparison uses detector-space peak positions, local line shapes, relative Bragg intensities, and projected Q_z profiles. Prior work has established the main ingredients needed for such a framework, including grazing-incidence scattering formalisms,[37, 38] reciprocal-space mapping and indexing for oriented thin films,[15, 17, 19, 23] and forward calculations of diffraction from textured films.[39] Complementary atomistic tools have also been developed for constructing commensurate multilayer 2D heterostructure supercells and visualizing reciprocal-space diffraction signatures, including Nookiin.[40] In contrast, the present work carries the calculation in detector space through detector geometry, grazing-incidence propagation, beam phase space, mosaicity, resolution, and ordered structure-factor weighting. The main-text validation is centered on the ordered 2D-powder limit using Bi_2Se_3 and Bi_2Te_3 films.[5, 7] Diffuse PbI_2 and stacking-disorder refinements are treated as later extensions of the same framework rather than as the primary proof of the method.

2 Diffraction Model

The detector-space forward model redistributes the structure-factor intensity of the contributing reflections over mosaic distributions. We Monte Carlo sample the incident beam profile across those orientations, weights each ray by transmission, and maps the summed intensity to detector pixels. The calculated and measured images are then compared using the same peak positions, line shapes, relative Bragg intensities, and projected L-integrated profiles.

2.1 Diffraction Geometry

We first consider a single incident ray striking a perfectly aligned crystallite, with no beam divergence, no refraction, and no absorption. Elastic scattering requires

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}, \qquad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i.$$

Diffraction occurs when the momentum transfer matches a reciprocal-lattice vector, $\mathbf{Q} = \mathbf{G}$. In the Ewald construction, this means that the reciprocal-space point for that reflection lies on the Ewald sphere of radius $|\mathbf{k}_i|$, and the simplest diffracted ray is obtained from

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}.$$

For a fixed $|\mathbf{G}|$, the set of possible reflection orientations forms a Bragg sphere of radius $|\mathbf{G}|$ about the reciprocal-space origin. The allowed exit vectors are found by intersecting that Bragg sphere with the Ewald sphere and then propagating the resulting exit ray to the detector.

For each trial incident ray, the model computes the sample intersection, applies the incidence geometry and optical corrections, finds the reciprocal-space intersections that satisfy elastic scattering, and propagates the resulting exit rays to the detector. Rays are Monte Carlo sampled from the probability distribution formed by the shape, bandwidth, and divergence of the beam. From every Bragg peak's structure factor spread upon the mosaic, a second Monte Carlo sampling occurs for each ray to choose an intersection point. Footprint, detector tilt, refraction, and absorption enter as experimental resolution and intensity terms. The coordinate conventions for the lab frame, detector plane, sample rotation, and illuminated footprint are summarized in Figs. 2 and 3.

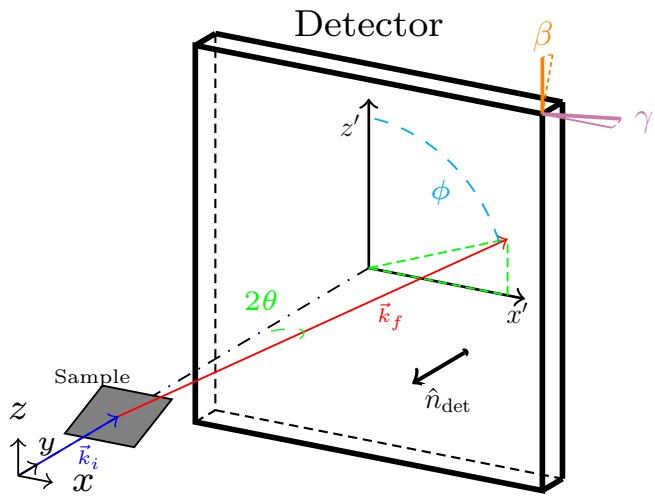

Figure 2: Overall lab-frame geometry. The incident wavevector $\mathbf{k}_i$ strikes the sample, and a representative exit wavevector $\mathbf{k}_f$ reaches the detector at scattering angle 2θ and detector-plane azimuth ϕ. The detector at distance D is tilted by β and γ.

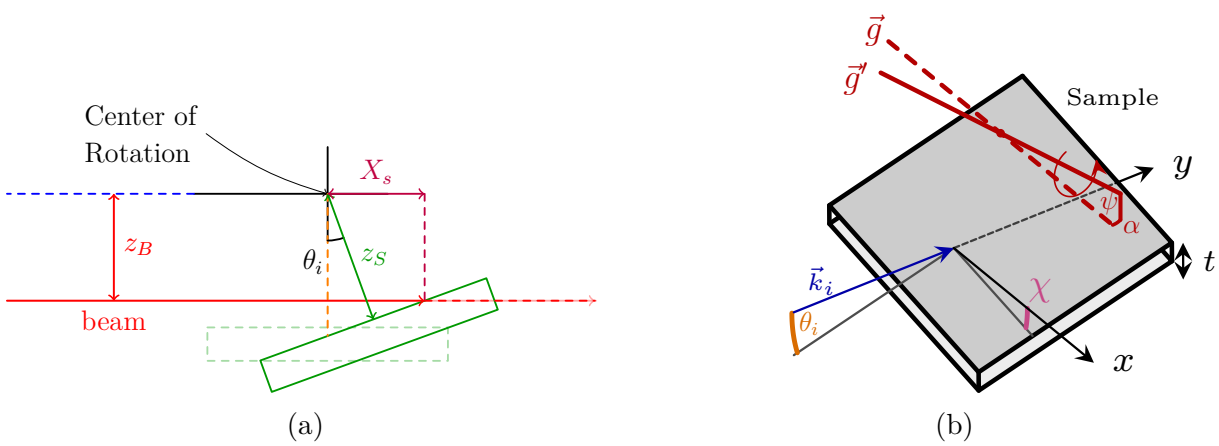

Figure 3: Two-stage sample geometry. Left: rotation about the goniometer center with sample offset z_S and beam offset z_B. θ_i is the incidence angle. Right: full alignment adds tilt δ and footprint length X_s within an illuminated window (W, H) on a film of thickness t. $\vec{g}$ is the goniometer arm (normal to $\hat{y}$), and $\vec{g}'$ is the misaligned arm obtained by rotations α and ψ.

2.2 Structure Factor

We enumerate integer in-plane Miller pairs (H, K), construct the reciprocal-space rod along G_z for each pair, and evaluate its detector intersections. For labels and profile extraction,

the in-plane pair is represented by the scalar family index m, defined by the in-plane sum

$$m(H,K) = \begin{bmatrix} H & K \end{bmatrix} \mathbf{M}_{\parallel} \begin{bmatrix} H \\ K \end{bmatrix}.$$

Here $\mathbf{M}_{\parallel}$ is the in-plane reciprocal-lattice metric, with any lattice-dependent normalization absorbed into the definition of m. For the hexagonal bismuth chalcogenides, this scalar is proportional to $H^2 + H\,K + K^2$. Thus, m labels an in-plane reciprocal-space rod family. In 2D-powder calculations, all integer (H,K) rods with the same m, and therefore the same Q_r, are evaluated individually and then summed after projection. Multiplicity and orientation effects therefore emerge within the forward model.

For each reflection with reciprocal-lattice vector

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*,$$

we compute the structure factor as

$$F_{\mathbf{G}} = \sum_a o_a\, f_a(Q)\, \exp[2\pi i\, \mathbf{G}\cdot\mathbf{r}_a]\, \exp\Big[-2\pi^2\, \mathbf{G}\cdot\mathbf{U}_a\cdot\mathbf{G}\Big], \tag{1}$$

where the sum runs over atoms a in the unit cell. Here o_a, $f_a(Q)$, $\mathbf{r}_a$, and $\mathbf{U}_a$ are the occupancy, atomic form factor, fractional position, and anisotropic displacement tensor for atom a. Symmetry constraints are retained during refinement, so only the independent crystallographic parameters are varied. The detector weight for each allowed ray is multiplied by $|F_{\mathbf{G}}|^2$ before mosaic and experimental broadening are applied.

2.3 Mosaicity

Mosaicity is the out-of-plane orientational spread of the layered crystallites about the mean film normal. It is distinct from the random in-plane rotation that defines the 2D-powder limit. Random in-plane rotation generates the equivalent m-indexed ring or rod family, while mosaicity determines how much of that reciprocal-space manifold is brought into the Ewald condition for a fixed incident angle.

We will see that the ordered-film images will require a two-component angular density,

$$\omega_{\mathrm{w}}(\Delta\theta;\eta,\Gamma,\sigma) = \eta L_{\mathrm{w}}(\Delta\theta;\Gamma) + (1-\eta)G_{\mathrm{w}}(\Delta\theta;\sigma), \qquad 0 \le \eta \le 1. \tag{2}$$

Here G_{w} is the Gaussian core, L_{w} is the Lorentzian tail, η is the Lorentzian weight, and $\Delta\theta$ is the crystallite tilt relative to the mean layer normal. The two widths are independent: the Lorentzian full width at half maximum is 2Γ, while the Gaussian full width at half maximum is $2\sqrt{2\ln 2}\,\sigma$. The wrapped form conserves probability over one angular period and avoids an artificial discontinuity at $\Delta\theta = \pm\pi$.

Figure 4 summarizes how the same mosaic distribution appears in reciprocal space and

on the detector. For $G_r \neq 0$, the redistributed intensity appears as off-specular band while for $G_r = 0$ it appears as cap-like.

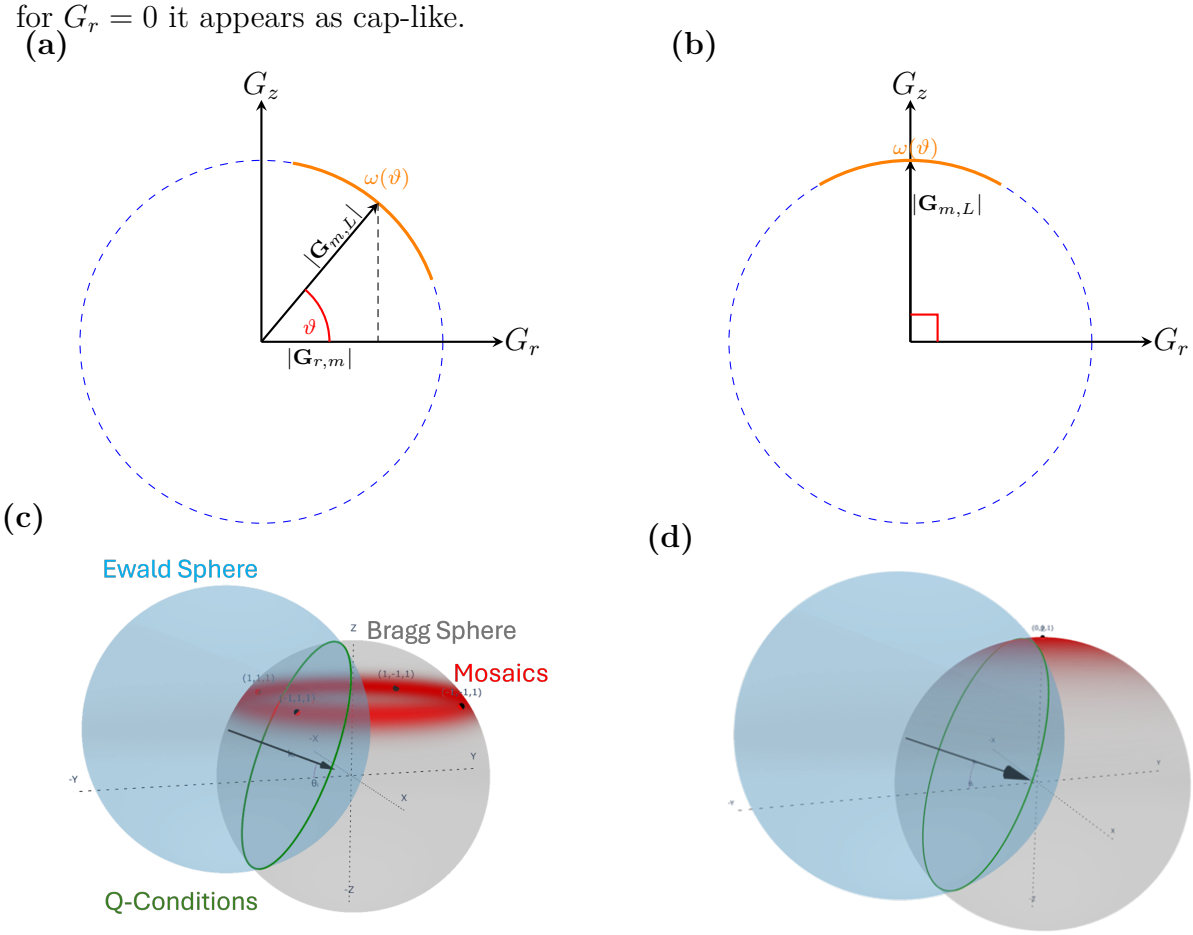

Figure 4: Mosaic broadening in reciprocal space and detector space. (a,c) For $G_r \neq 0$, random in-plane rotation and finite out-of-plane mosaicity give off-specular arcs. (b,d) For $G_r = 0$, the same orientational distribution gives cap-like $00L$ intensity near the specular direction. The narrow Gaussian-like core controls the main width of near-condition features, while the Lorentzian-like tail supplies the weak tilted-crystallite population needed for off-condition $m = 0$ peaks.

3 Refinement workflow

We use a staged refinement because geometry, mosaicity, and crystallographic intensity are strongly correlated in a fully global fit. Each stage uses the detector-space observable most sensitive to one parameter group before the next group is introduced. Table 1 summarizes the ordered-film workflow used for Bi_2Se_3 and Bi_2Te_3.

The instrumental and alignment stages fix where an ideal reflection should appear on the detector. Direct-beam and calibrant data define the beam phase space and detector mapping, while indexed sample reflections determine the incidence condition, sample offsets, and goniometer misalignment.

Table 1: Staged refinement workflow used for the ordered Bi_2Se_3 and Bi_2Te_3 films. All sample images are dark-subtracted before the listed observables are formed. Diffuse-scattering and stacking-disorder refinements are deferred from the present main-text draft.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x$, $\Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D, detector tilts (β, γ), and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry $(\theta_i, \delta, z_B, z_S)$, in-plane rotation χ, goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected local detector ROIs, specular-tail checks, and local profile shapes	Gaussian-plus-Lorentzian mosaic parameters	Center-aligned profile shapes, off-specular arcs, and specular tails
5	Ordered-film validation	Selected detector-space Bragg regions and Q_z projections from fixed-m trajectories	Image scale factors and ordered structure-factor parameters	Relative Bragg intensities and measured/calculated line-shape agreement

The remaining stages refine peak shape and intensity. Local detector profiles constrain the shared Gaussian-plus-Lorentzian mosaic distribution. Selected Bragg-region integrals constrain ordered structure-factor parameters, and the final validation projects measured and calculated images along the same fixed-m trajectories for comparison versus Q_z. Implementation details required for efficiency, including 2θ–ϕ remapping, caked fields, lookup tables, and Monte Carlo sampling, are described in the supporting information. Later stacking disorder can be added by fitting the m-indexed rods as a function of L.

4 Results: ordered Bi_2Se_3 and Bi_2Te_3 films

Ordered Bi_2Se_3 and Bi_2Te_3 films validate the model by showing that the same detector-space forward model quantitatively reproduces measured line shapes and intensities in two related layered chalcogenides. Three detector images, collected at incident angles of 5°, 10°, and 15°, were used in the refinement. Figure 5 shows the 5° images and the associated projected profiles because this incidence condition gives the clearest single-view display of both the off-specular arc families and the specular $HK = 0$ features. Together, the data test the 2D-powder orientational limit, sample geometry, finite beam phase space, mosaicity, detector resolution, and ordered structure-factor weighting.

The figure is meant to be read as both a validation plot and a result figure. The off-specular $HK = 1$, 3, and 4 profiles show that the calculated 2D-powder geometry places the arc intersections at the correct detector positions and reproduces their L-dependent line shapes. More unexpectedly, the same images contain strong specular-family intensity from $HK = 0$ reflections that are not all selected by the nominal incidence condition. In a narrowly aligned film with only the central incidence angle and a rapidly decaying mosaic distribution, only a small subset of the $00L$ family would carry appreciable intensity. The

measured $HK = 0$ profiles instead show additional $00L$ peaks, including weak off-condition peaks that remain visible on a semilogarithmic scale. These peaks provide the most direct evidence that the mosaic distribution has a long tail rather than only a narrow Gaussian-like core.

A second nontrivial observation is the bright low-L intensity near the base of the $HK = 0$ trajectory in the detector images. This intensity lies close to the direct-beam, beamstop, and critical-angle region and is not expected from an ideal narrow-band Bragg selection of isolated $00L$ peaks. It is therefore not treated as ordinary reflectivity in the ordered-film comparison. Instead, it is used as a qualitative check on whether the forward model includes the relevant experimental phase space: finite wavelength bandwidth, finite angular divergence, sample mosaicity, and the effective thickness of the Ewald shell. Because the low-L Bragg sphere is small in reciprocal-space radius, those finite-width effects illuminate a comparatively large angular fraction of the near-origin specular region and can produce the observed star-like detector morphology.

The detector-space panels and profile panels in Fig. 5 use the same extraction convention. In panels (a) and (b), the solid colored curves are the calculated centerlines of selected fixed-Q_r/HK Bragg-rod trajectories on the detector. The semi-transparent colored bands are the finite detector regions integrated to form the one-dimensional profiles. The dashed colored outlines mark the edges of those integration bands; they are not additional reflections. The central blue band is the $HK = 0$ specular-family trajectory, while the paired off-specular bands are the two detector-side intersections of the same 2D-powder rod family. The plus and minus labels identify those two branches only; they do not denote different crystallographic families. Each HK label denotes a family of symmetry-equivalent rods with the same Q_r, which are generated and projected separately before their detector-space contributions are summed.

Panels (c) and (d) show the corresponding line profiles after applying the same projection and integration operation to the measured and calculated images. The horizontal coordinate is $L = Q_z/|\boldsymbol{c}^*|$, so a marker labeled $L = n$ denotes the nominal $00n$ or HKn Bragg condition for that selected rod family. The black solid curves are the measured profiles. The orange dashed curves are the simulated profiles after the same detector-space projection. Open circles mark the indexed Bragg/fit positions used to orient the comparison. The off-specular rows are plotted on a linear intensity scale because they test peak positions, relative intensities, and arc widths. The $HK = 0$ row is plotted on a logarithmic scale because the important result is the persistence of weak, unexpected $00L$ intensity far from the strongest specular peak.

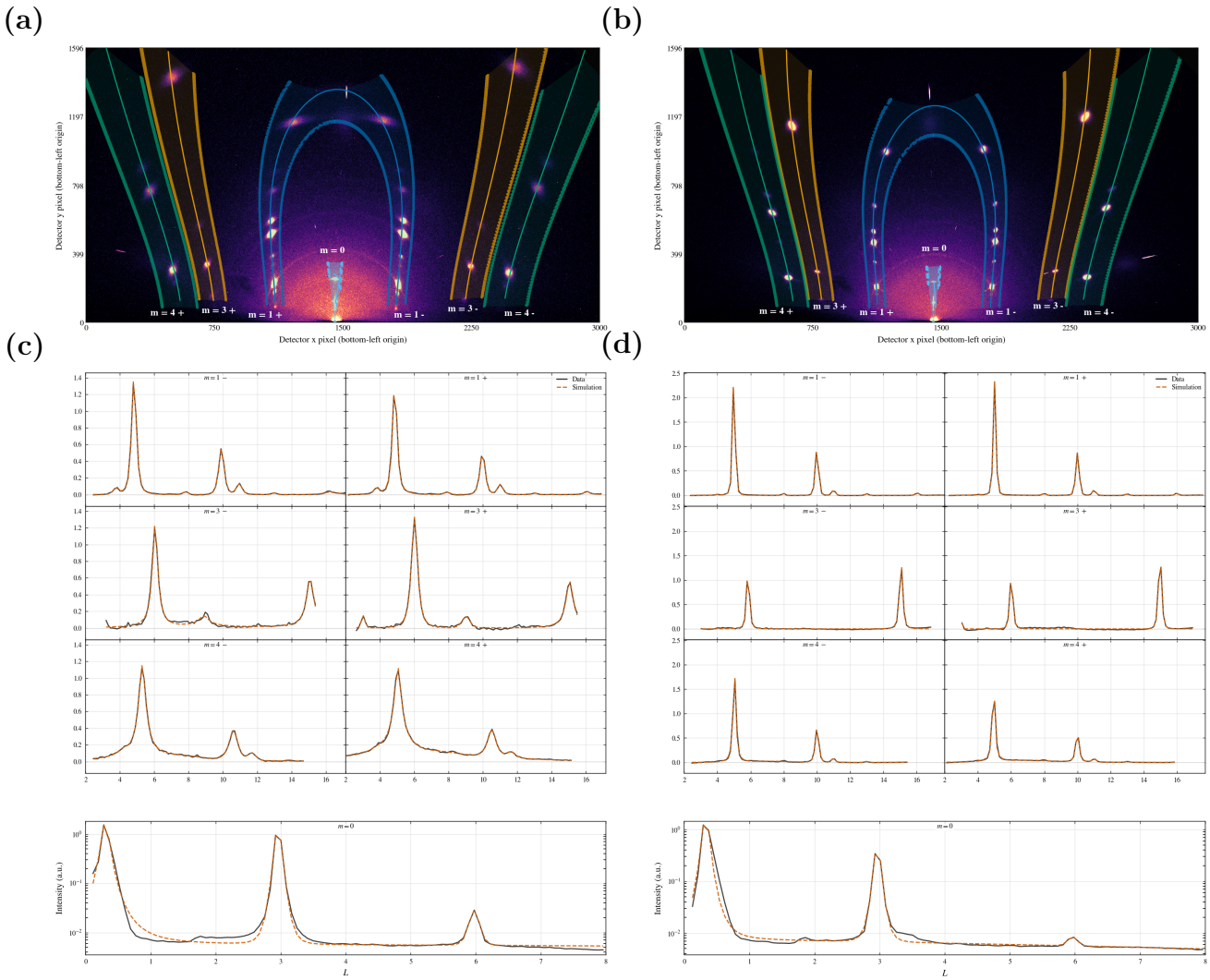

Figure 5: Ordered-film validation of Bi_2Se_3 and Bi_2Te_3 at $\theta_i = 5°$. (a,b) Measured detector images with selected Q-space extraction regions overlaid. Solid colored curves are the trajectory centerlines. Semi-transparent colored regions are the detector pixels integrated to form the profiles. Dashed colored outlines mark the integration-band boundaries. The labels m identify 2D-powder rod families, and the signs label the two detector-side branches of the same family. The central $m = 0$ band follows the specular family and includes the bright low-L intensity near the critical-angle region. (c,d) Extracted profiles plotted versus L. Black solid curves are measured data and orange dashed curves are simulations after the same projection. The off-specular rows test the detector geometry, 2D-powder construction, mosaic-core width, and ordered structure-factor pattern. The semilogarithmic $m = 0$ rows emphasize the unexpected persistence of additional $00L$ peaks, which requires a long Lorentzian mosaic tail in addition to the narrow mosaic core.

Draft figure placeholder

`figures/results_ordered/hk0_l3_star.png`

Close-up of the low-L $HK = 0$ detector feature. Show the star-like morphology and the bright near-origin intensity above the beamstop/critical-angle region with minimal annotations.

Figure 6: Close-up of the low-L $HK = 0$ detector feature. The bright near-origin intensity and vertical star-like morphology are not expected from an ideal, narrow-band calculation containing only isolated on-condition $00L$ peaks. In the model interpretation, the feature reflects the finite experimental phase space: wavelength bandwidth and angular divergence give the Ewald sphere an effective thickness, while the long mosaic tail supplies tilted crystallites that bring otherwise off-condition specular-family intensity into the measured detector region.

Because mosaicity, beam divergence, wavelength spread, incidence-angle uncertainty, detector resolution, and finite integration width all broaden the effective scattering condition, the projected coordinate should not be interpreted as an ideal reciprocal-space cut from a single perfect crystallite. The comparison remains meaningful because the calculated image is subjected to the same projection and resolution effects before it is compared with the measured profile. The figure therefore tests the detector-space calculation as used experimentally.

In Fig. 5, the off-specular and $HK = 0$ profiles play complementary roles. The off-specular arcs check the 2D-powder reciprocal-space construction and calibrated detector geometry, while their widths and asymmetries constrain the Gaussian-like mosaic core and instrumental resolution. The off-condition $HK = 0$ peaks test the Lorentzian mosaic tail and show why a rapidly decaying mosaic core alone is insufficient. Figure 6 isolates the low-L specular-region feature, emphasizing that the ordered chalcogenide images contain more information than peak positions alone: the unexpected $00L$ peaks and near-origin specular intensity both require the calculation to include the finite beam phase space and the tails of the out-of-plane orientation distribution.

5 Discussion and conclusion

The ordered Bi_2Se_3 and Bi_2Te_3 films validate the mosaic model and detector-space forward model. A single calculation accounts for detector geometry, finite beam phase space, sample alignment, mosaicity, resolution, and ordered structure-factor weighting, then compares measured and calculated Q_z profiles along the same fixed-Q_r/HK trajectories. The key result is direct line-shape agreement.

Keeping the comparison in detector space is essential because the constraining intensity is distributed among off-specular arcs, cap-like $HK = 0$ features, weak off-condition specular peaks, and bright low-L specular-region intensity that a conventional 1D reduction would obscure. Because the calculated image includes the same effects and is projected in the same way as the measurement, the overlay tests the complete experimental model. The off-condition $HK = 0$ peaks show why a narrow Gaussian-like mosaic core is insufficient and why the Lorentzian tail is required, while the near-origin low-L feature emphasizes the importance of finite beam and wavelength phase space.

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